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Introduction to SPECviewperf

The SPECviewperf® 15 benchmark, developed by the SPEC® Graphics & Workstation Performance Group's ([SPECgwpwg](#)) Graphics Performance Characterization (SPECgpc®) subcommittee, is the worldwide standard for measuring graphics performance based on professional applications. The benchmark measures the 3D graphics performance of systems running under the OpenGL, DirectX, and Vulkan application programming interfaces. The benchmark workloads represent graphics content and behavior from actual workstation-class applications, without installing the applications themselves:

- *3dsmax-08*: derived from Autodesk 3ds Max 2023 using DirectX 11
- *blender-01*: derived from Blender 3.6 LTS using OpenGL
- *catia-07*: derived from Dassault Systèmes' CATIA V5 and 3DEXPERIENCE CATIA applications using OpenGL
- *creo-04*: created from PTC's Creo 9™ using OpenGL
- *energy-04*: inspired by OpendTect seismic visualization, uses OpenGL
- *enscape-01*: built on Chaos's Enscape 4.0 using Vulkan with raytracing
- *maya-07*: created with Autodesk Maya 2025 using OpenGL
- *medical-04*: exploring datasets from a beating heart, brain, and alligator using the Tuvok library and OpenGL
- *snx-05*: created from Siemens NX 2406 using OpenGL
- *solidworks-08*: derived from Dassault Systèmes' Solidworks 2024 using OpenGL
- *unreal_engine-01*: built on Unreal Engine 5.4.2 using DirectX 12 with raytracing

For more information, see [benchmark page](#).

System Requirements

Minimum System Specifications

The following is the minimum supported system configuration. Some workloads may still run on systems that do not meet these requirements, but support will not be provided by SPEC for any such configuration.

Component	Requirement
OS	<i>Windows 11 Pro version 23H2</i>
CPU	<i>Intel 12th Gen or AMD Ryzen 7000 series or newer</i>
Graphics	<i>AMD, Intel, or NVIDIA GPU with 4GB or greater shared or dedicated memory, OpenGL 4.5, DirectX 12, Vulkan 1.1 and Hardware H.265 encoding</i> See Workload Requirements
RAM	<i>15GB available system RAM and at least 1GB/thread</i>
Disk Space	<i>At least 100GB free space for install and execution</i>
Display Resolution	Minimum 1920 x 1080

Note:

- The benchmark run requirements are the minimum requirements that we guarantee the benchmark will run with. This is not the same as the submission requirements, as you can meet the run requirements without qualifying as a [submission candidate](#).

Calculation of SPECviewperf15 Memory Requirements

In the benchmark, memory is categorized based on its allocation to GPU and CPU, and is defined differently for discrete and integrated GPU setups. Here's how memory is calculated for each scenario:

System Type	VP_GPU_memory	VP_CPU_memory
Discrete GPU Systems	Dedicated GPU memory	Total physical CPU memory available
Integrated GPU Systems	Dedicated GPU memory + Shared GPU memory	Total physical CPU memory available minus the larger of: (1) Memory already dedicated to GPU or (2) 4GB (for optimal GPU performance)
General Requirements	Must be at least 4GB	Must be at least 15GB

Note:

- These requirements are as per Full HD resolution.
- When running the workloads in 4K resolution, the application will require more memory. Running on systems with limited resources may lead to timeouts, system hangs, or graphical corruptions.

Workload Specific Requirements

Some workloads test features that require special hardware and driver support. If your GPU does not meet these requirements, those workloads will be disabled.

Workload	Requirement(s)
catia-07	<i>at least 6GB video memory</i>
energy-04	<i>at least 4GB video memory</i>
enscape-01	<i>Hardware Ray Tracing Support; at least 6GB video memory</i>
unreal_engine-01	<i>An AMD, Intel, or NVIDIA graphics card that supports DirectX 12 (with Shader Model 6.6 atomics), hardware ray tracing (DXR Tier 1.1 and Resource Binding Tier 3); at least 8 GB of video memory.</i>

Note:

- When running certain workloads, entry-level graphics and older hardware may exhibit a lock-up as a result of a Windows Timeout Detection and Recovery (TDR) event. Refer to [Microsoft's documentation on TDR](#) and your hardware documentation for instructions on how to increase the TDR timeout threshold (TdrDelay of at least 10) or disable TDR detection (TdrLevel=0) with TDR registry settings.

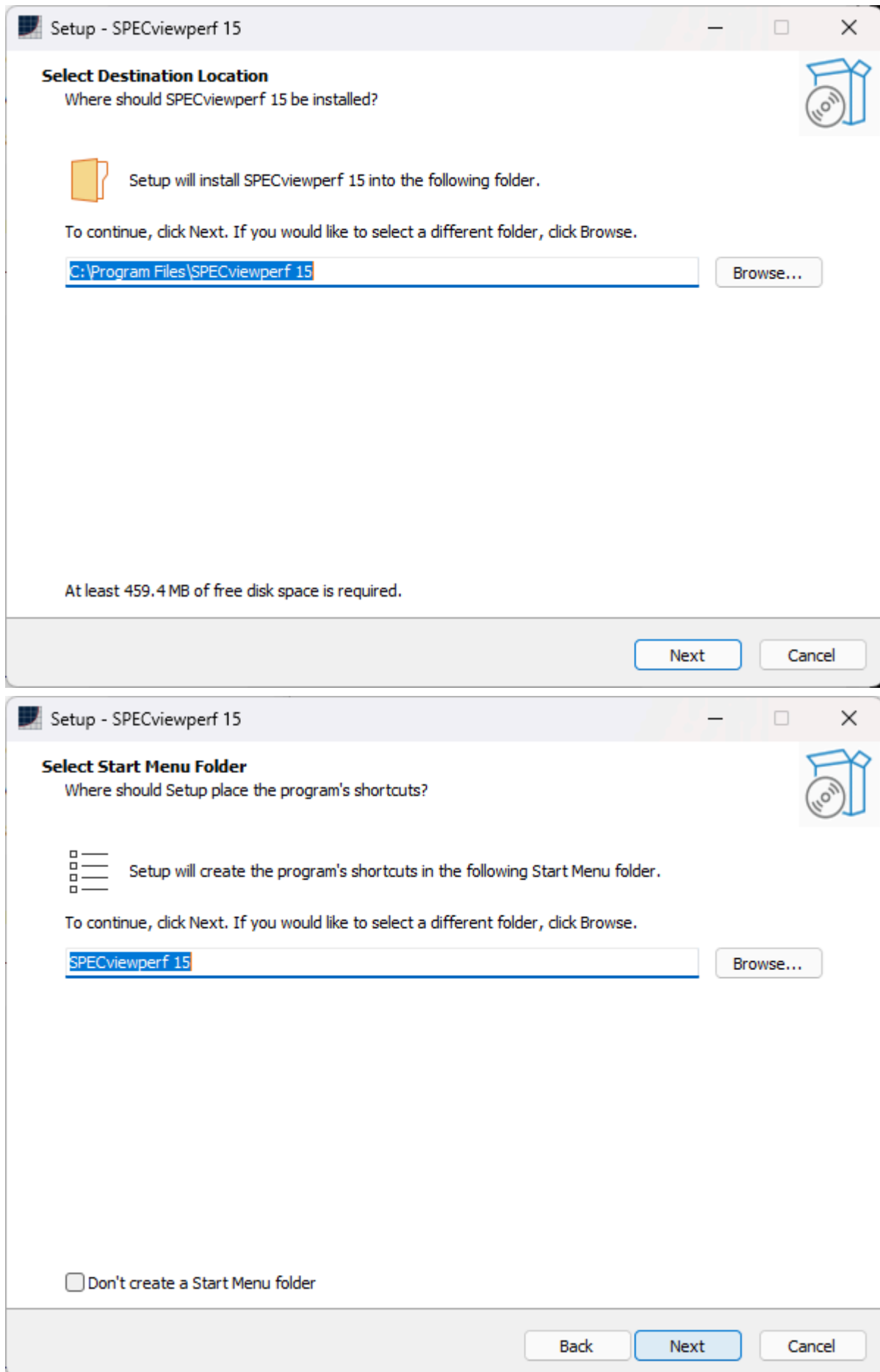
Installation Guide

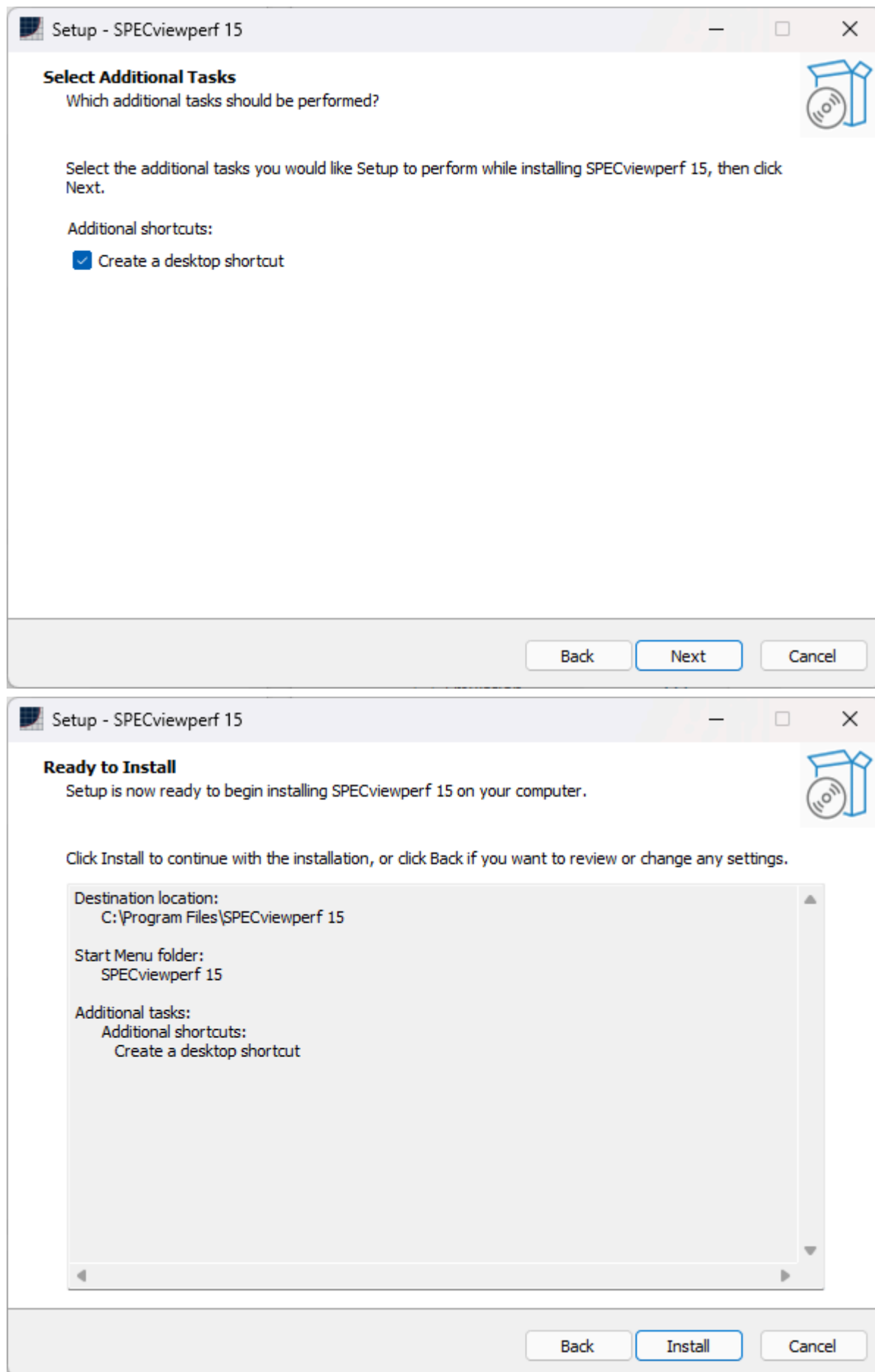
Downloading SPECviewperf

The SPECviewperf 15 benchmark can be downloaded from the [Download](#) tab of the [SPECviewperf15 benchmark page](#). It is free to download for everyone except sellers of computers and related products. A paid license is required for any for-profit entity that sells computers or computer related products in the commercial marketplace, with the exception of [SPECgwp member companies](#).

Step-by-Step Installation Instructions

1. Once you have downloaded the installation package for SPECviewperf 15, run the installer. It will give you an option of where to install the benchmark - the default install location is "C:\Program Files\SPECviewperf 15".

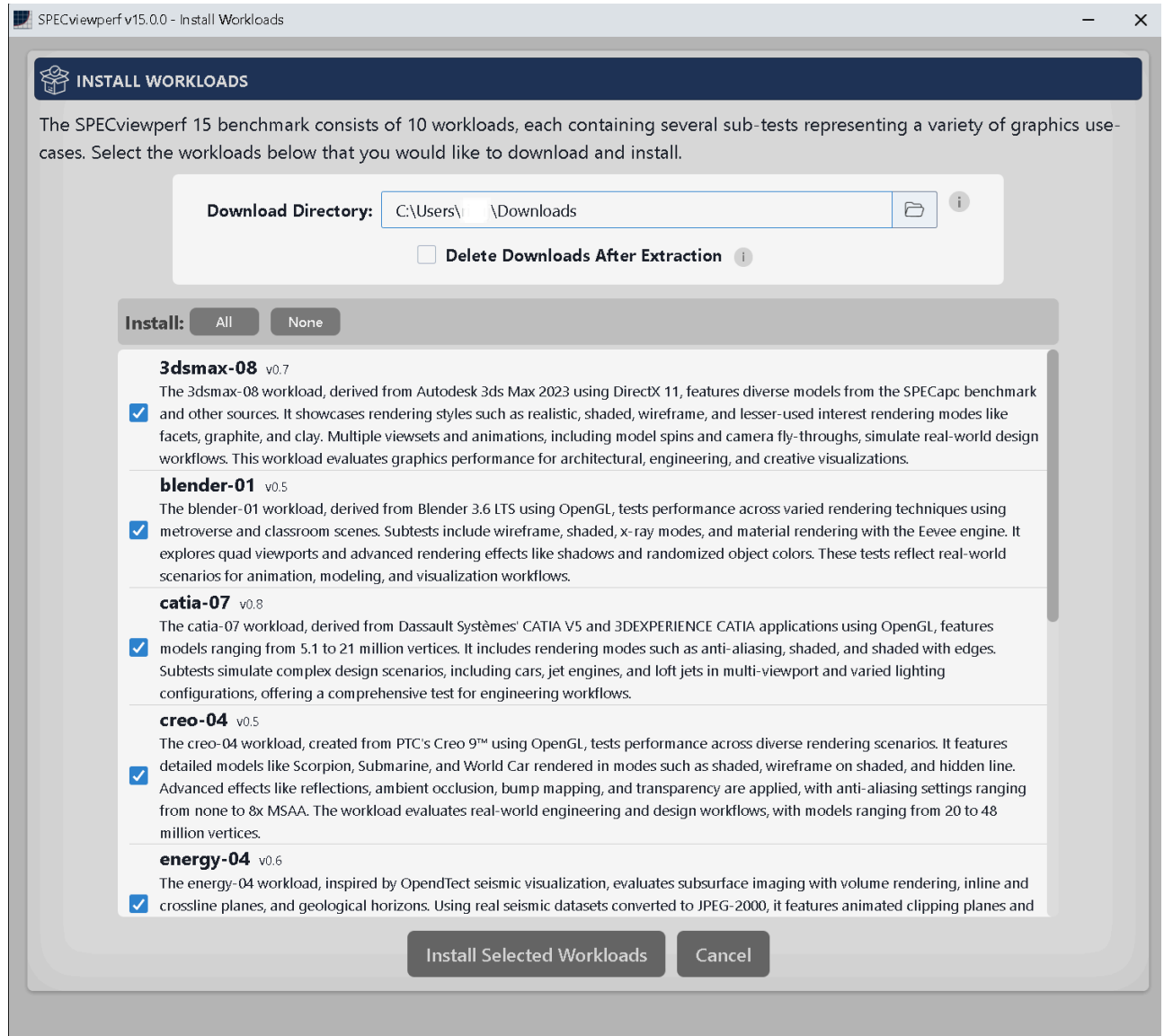




- After the initial installation is complete, you will be presented with a dialog asking you which workloads you wish to download and install. By default, all official SPECviewperf 15 workloads are selected; however,

you may choose any subset of them you like. The benchmark requires at least one workload to be installed to run.

Note: By default, the downloaded workload packages are deleted after extraction. If you un-check "Delete Downloads After Extraction" the workload packages will be retained in the download location you select and you can use them later for an off-line installation on the same or a different system.



Offline Install

To create an offline install package, users must first install the benchmark on a system with an internet connection, unselect the **Delete Downloads After Extraction** option, and select all workloads to be included in the offline install. The downloaded workload packages should be placed in the same folder as the installer and may be copied to other machines for offline installation.

Benchmark Usage

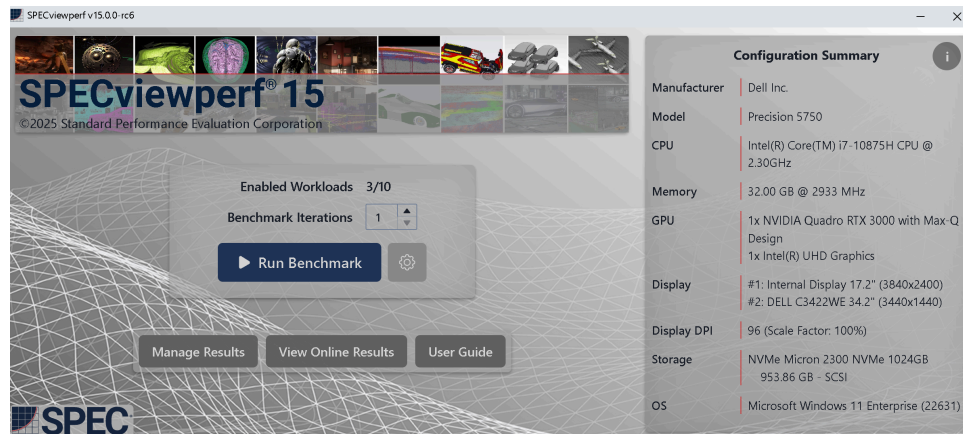
Preparing Your System for Benchmarking

Refer to the [run rules](#) for benchmark compliance prior to running SPECviewperf.

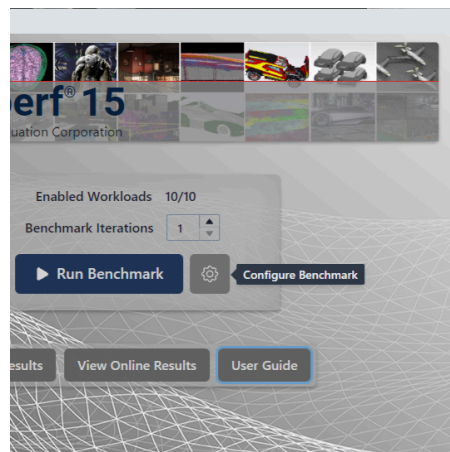
Main Window

The main GUI interface allows the user to configure and run the benchmark.

- The **Enabled Workloads** field shows how many workloads have been selected to run out of the total number of workloads available. The value will be shown in red font if the run will not be considered a **submission candidate** or errors are detected with any workload. Opening the **benchmark configuration** window may provide additional details on workload errors.
- Workload selection can be changed via the benchmark configuration menu opened with the gear icon next to **Run Benchmark** button.



SPECviewperf graphical user interface (GUI)

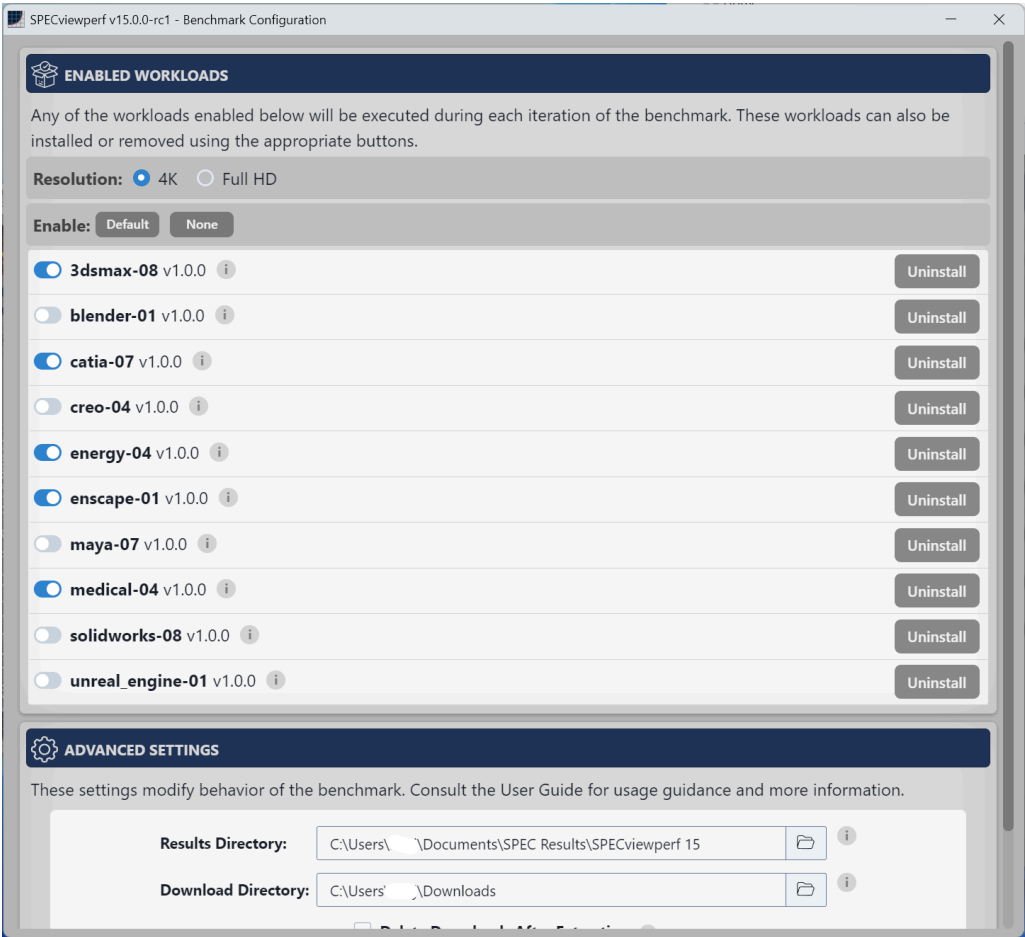


SPECviewperf GUI Configure Benchmark Button

Benchmark Configuration

The benchmark configuration window allows users to select which workloads to run. The benchmark configuration window will also present options to install/uninstall a workload, or update a workload if a new version is available.

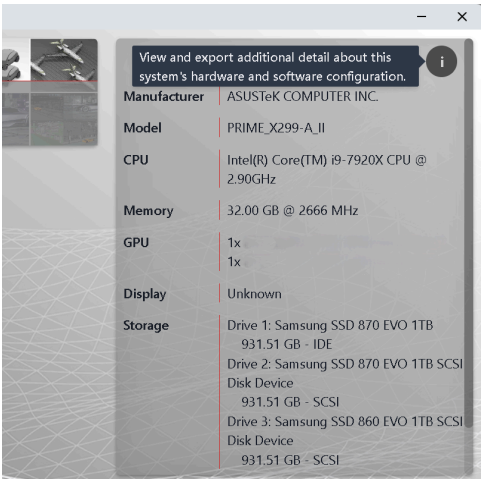
The user can also choose the resolution they want to run the workloads at via the radio buttons on the top of the window. The default is **4K** (3840x2160) if the display supports it, else it falls back to **Full HD** (1920x1080).



Benchmark configuration window for workload selection

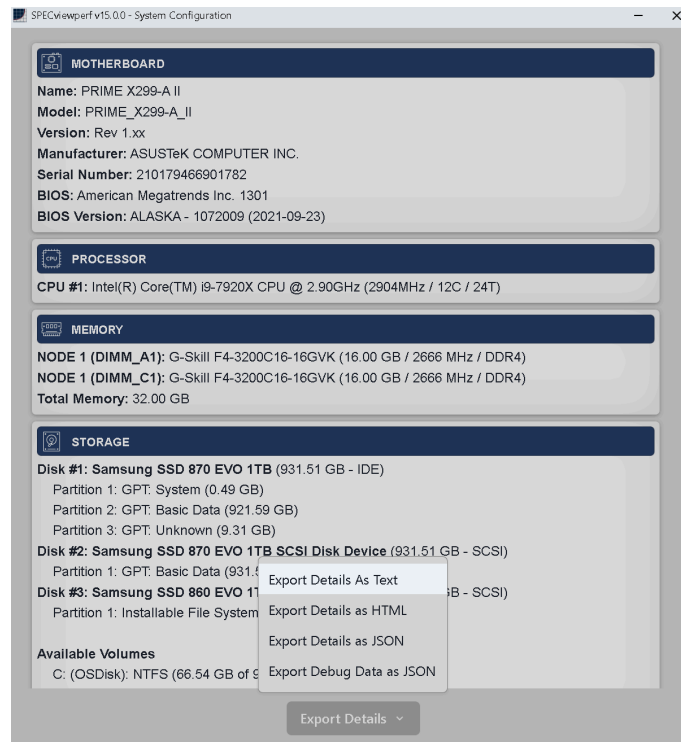
System Configuration Report

System configuration details collected by the benchmark can be viewed by clicking the info icon in the top right of the [main window](#).



Info icon to open system configuration window

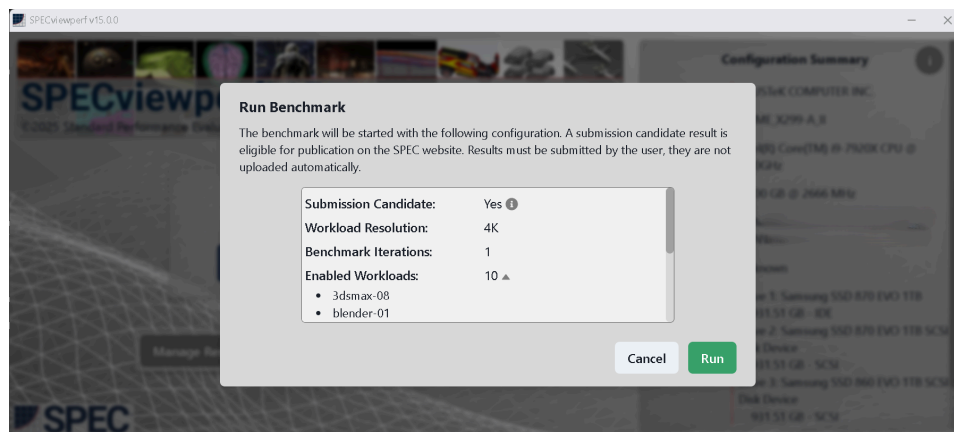
Options are provided to export details in various formats. Note that the detected configuration details are embedded into all result files. System details do not contain user-identifiable information and users can verify no personal details are exposed by inspecting the information in this screen. The **Exporting Debug Data** option may generate output containing identifiable information, but is typically only needed for debugging and support requests.



System configuration Details

Running the Benchmark

Selecting **Run Benchmark** from the main window will display a confirmation window listing the benchmark configuration including workloads, resolution and iterations for the run. The confirmation window will also state if the run can generate a [result for submission](#) to SPEC to publish on the official [SPECviewperf results](#) page.

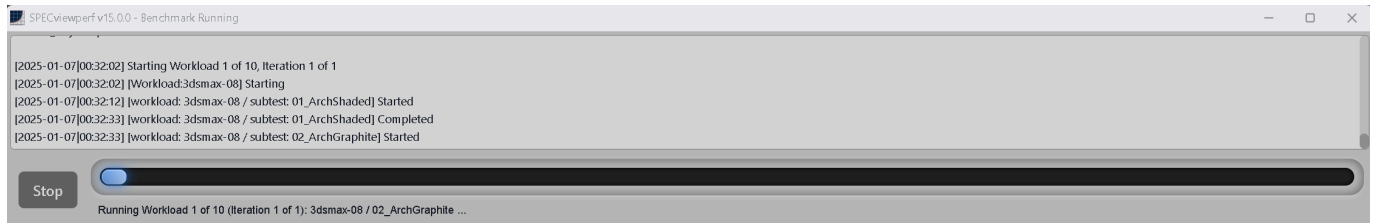


Run benchmark confirmation window

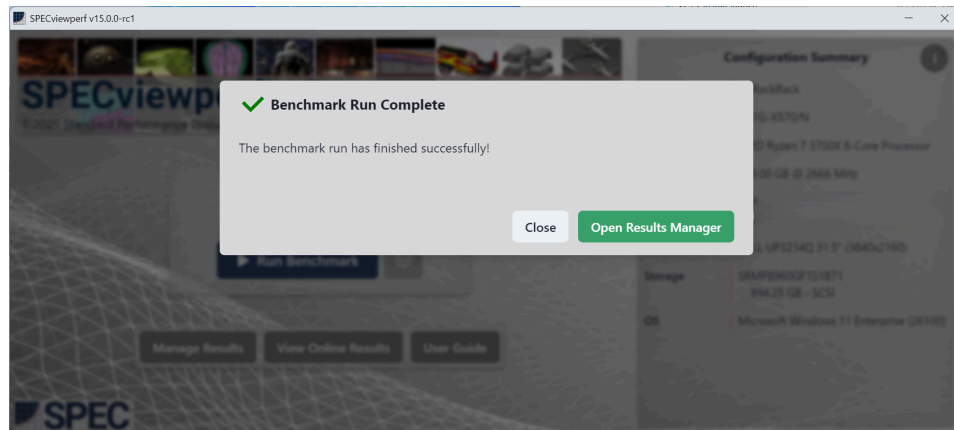
Note: the benchmark does not automatically send or upload any information to SPEC. Results or system configuration details are not uploaded or collected automatically -- they must be properly [packaged](#) and sent to SPEC by the user.

Run Status and Completion

During a run, the benchmark status window will show which workload and subtest is currently executing and the overall progress of the benchmark run. If a user wishes to stop the benchmark during a run, it can be done from this window. Upon pressing the **Stop** button, the GUI will then pop up a confirmation window to stop or resume the benchmark execution.

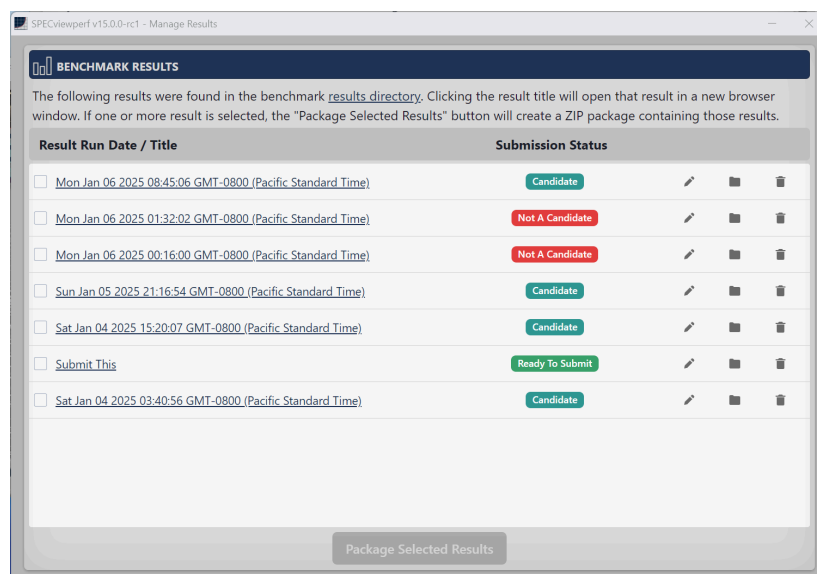
*Benchmark status window*

After a run completes, the user is given the option to open the results manager.

*Run complete window*

Manage Results

The **Manage Results** button in the **main window** or the **Open Results Manager** button in the **run completion window** opens the results manager. Clicking a result will open the HTML result file.

*Manage results window*

The **results directory** link at the top of the results manager will open the currently set result folder. The default result folder is %USERPROFILE%\Documents\SPEC Results\SPECviewperf 15, but can be changed from the **advanced settings menu**. Note that results may be marked **Candidate** or **Not A Candidate**. If the submission status is **Candidate**, the results may be **packaged and submitted** to SPEC for review and publication. If submission details are complete, the submission status will show **Ready To Submit**.

A result's directory may be accessed by clicking the folder icon next to a result. The result directory includes results in HTML, CSV, and JSON formats, along with logs, screen grabs, and other meta-data from each selected workload from the run. If a result is expected to be submitted to SPEC for publication, all files in the result directory should remain unaltered.

Results may be annotated with a title and other submission information. Clicking the pencil icon in the result row will open the results / submission details window.

The screenshot shows a window titled "Result / Submission Details" with the following fields:

- Result Title:** Enter a short description for this configuration...
- Run Date:** Mon Jan 06 2025 08:45:06 GMT-0800 (Pacific Standard Time)
- Benchmark Version:** SPECviewperf 15.0.0-rc1
- Submission Candidate:** Yes
- Submitter's Company Name:** Enter your company name...
- Submitter Name:** Enter your full name...
- Submitter E-mail:** Enter your e-mail address...
- Submission Comments:** Enter any comments related to these results. Please include any non-standard system settings...

Results / Submission details window

Adding submission details is required only if submitting a result to SPEC. Users may also use this functionality to rename results and keep notes for their own tracking purposes.

Result Packaging and Submission

The [Manage Results](#) window allows users to display results from all the benchmark runs for packaging and/or submission. Selecting one or more results with the checkbox will enable the [Package Selected Results](#) button, which generates a zip file containing the selected results. The result package will be created in the same folder as the results (default folder: `%USERPROFILE%\Documents\SPEC Results\SPECviewperf 15`).

The screenshot shows a dialog box titled "Package Results" with the following text:

This will create a ZIP package containing the selected result. This contains system configuration details, benchmark logs, and benchmark scores from each result.

☒ Use result title as directory name inside package.

Buttons: Cancel, Create Package

Select results to package from the result manager window

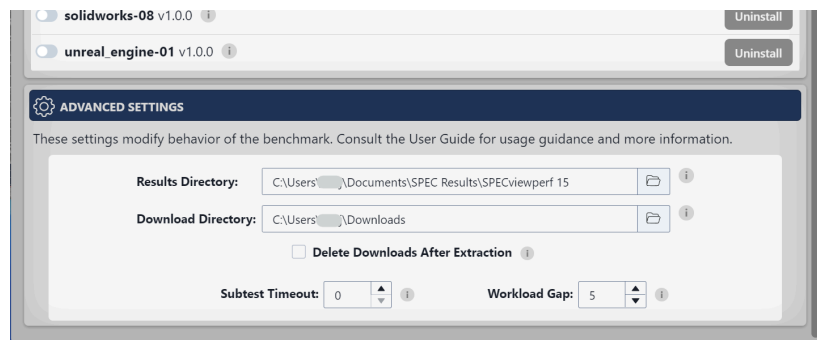
A user may package results for their own convenience or, if a result is marked as a **Candidate**, for submission to SPEC.

- Submissions to SPEC must include complete submission details for every packaged result.
- Submission comments must include any performance-relevant system customization details that are not captured in the [system configuration](#) window and may include any additional information helpful to the reviewer. Examples of additional information to include in the comments:
 1. Links to GPU/accelerator driver download web pages
 2. Any modified BIOS settings, e.g. disabling SMT or hyper-threading, overclocking, RAM speed/timings, etc.
 3. Windows power profile selection and any changes to default profiles
 4. Disabling VSYNC
 5. Any changes to the default setting of in Virtualization Based Security (Memory Integrity in Device Settings)
 6. Status of Windows Real-time protection and any virus scan exclusions
 7. Links to system specification or purchase web pages

Before submitting, review the complete [submission candidate](#) criteria.

Advanced Configuration Options

The advanced settings menu is located at the bottom of the [Benchmark Configuration](#) window. Each option is described in the table below.



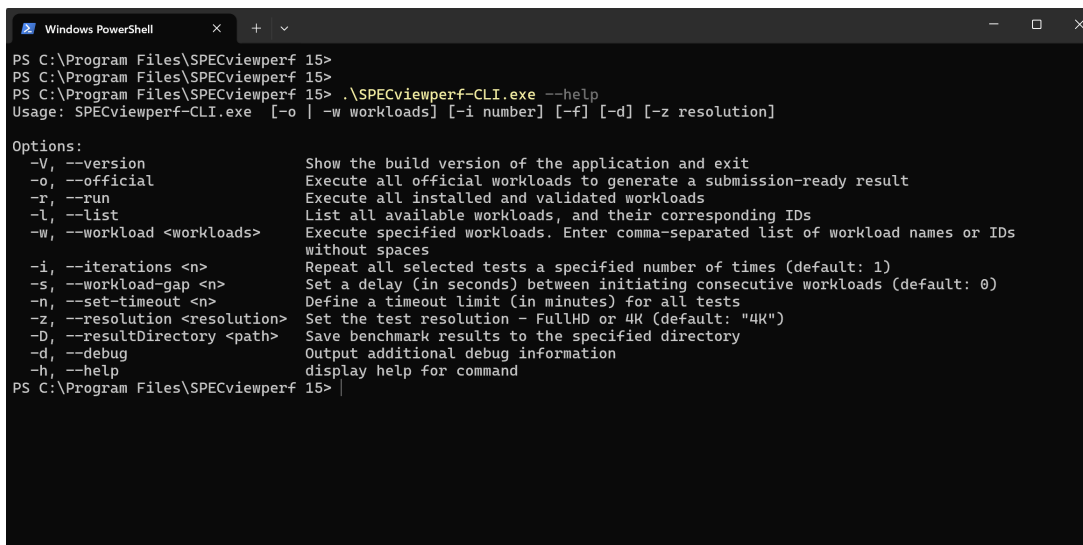
Advanced settings menu

Option	Description
Results Directory	Change the default results directory where results are saved. The result manager window will only show results from the directory listed in this field. If changed, the result directory can be reset to default by clicking the reset icon next to the input field.
Download Directory	Change the default directory where workload installation packages are downloaded and saved.
Delete Downloads After Extraction	If checked, automatically deletes workload installation packages after successful installation. Uncheck this if you plan to create an offline install bundle.
Subtest Timeout	If set to a non-zero value, tests will terminate after that defined timeout in minutes and proceed to the next test. Disable this if your system is encountering frequent timeouts and you believe the workload is still running correctly by setting to 0 (default.)

Option	Description
Workload Gap	Add a waiting period between each workload execution. May be used to allow systems to thermally recover between workloads.

Command-Line Interface

The command-line interface (CLI) enables easy automation and integration into scripts. The CLI is invoked with the **SPECviewperf-CLI.exe** binary found in the install folder (default **Program Files\SPECviewperf 15**). Run the CLI without any options or with the help command, **SPECviewperf-CLI.exe --help**, to display all available options.



```

Windows PowerShell
PS C:\Program Files\SPECviewperf 15>
PS C:\Program Files\SPECviewperf 15>
PS C:\Program Files\SPECviewperf 15> .\SPECviewperf-CLI.exe --help
Usage: SPECviewperf-CLI.exe [-o | -w workloads] [-i number] [-f] [-d] [-z resolution]

Options:
  -V, --version           Show the build version of the application and exit
  -o, --official          Execute all official workloads to generate a submission-ready result
  -r, --run               Execute all installed and validated workloads
  -l, --list              List all available workloads, and their corresponding IDs
  -w, --workload <workloads> Execute specified workloads. Enter comma-separated list of workload names or IDs
                           without spaces
  -i, --iterations <n>   Repeat all selected tests a specified number of times (default: 1)
  -s, --workload-gap <n> Set a delay (in seconds) between initiating consecutive workloads (default: 0)
  -n, --set-timeout <n>  Define a timeout limit (in minutes) for all tests
  -z, --resolution <resolution> Set the test resolution - FullHD or 4K (default: "4K")
  -D, --resultDirectory <path> Save benchmark results to the specified directory
  -d, --debug             Output additional debug information
  -h, --help              display help for command
PS C:\Program Files\SPECviewperf 15>

```

Command-line interface

Some common CLI options are shown in the following table:

CLI Option	Description
--help / -h	Display all options and parameters
--list / -l	List available workloads and subtests
--official / -o	Run all official workloads
--run / -r	Run all installed workloads
--workload / -w [workload-list]	Run selected workloads
--resolution / -z [resolution]	Run workloads at Full HD or 4K resolution

During a CLI run, the benchmark will accept the following commands to stop or skip workloads:

Command	Description
Ctrl+C	Cancel benchmark run, attempt to save partial results
x	Abort and skip the currently running workload

Example Usage:

1. Running Specific Workloads: To run selected workloads such as 3dsmax-08 and catia-07, use the following command:

```
> .\SPECviewperf-CLI.exe -w 3dsmax-08,catia-07
```

Output:

```
[2025-01-08|09:21:17] Gathering System Details: Started
[...]
[2025-01-08|09:21:32] Workload initialization: Finished

Valid Workloads:
  1) 3dsmax-08 [v1.0.0]
    ✓ 01_ArchShaded
    [...]
  3) catia-07 [v1.0.0]
    ✓ 01_FourICDCarsWhite
    [...]

[2025-01-08|09:21:32] Starting Workload 1 of 2, Iteration 1 of 1
[2025-01-08|09:21:32] [Workload:3dsmax-08] Starting
[...]
```

Notes:

- Replace 3dsmax-08,catia-07 with the desired workload names or IDs.
- Use --list (-l) to view all available workloads before running the command.

```
> .\SPECviewperf-CLI.exe -w 2,4 -z FullHD
```

Output:

```
[2025-01-08|09:46:48] Gathering System Details: Started
[...]
[2025-01-08|09:46:56] Workload initialization: Finished

Valid Workloads:
  2) blender-01 [v1.0.0]
    ✓ 01_MetroverseWireframeQuad
    [...]
    ✓ 07_ClassroomEevee

  4) creo-04 [v1.0.0]
    ✓ 01_ScorpionShadedReflections
    [...]
    ✓ 13_ScorpionNoHidden

[2025-01-08|09:46:56] Starting Workload 1 of 2, Iteration 1 of 1
[2025-01-08|09:46:56] [Workload:blender-01] Starting
[2025-01-08|09:46:56] [workload: blender-01 / subtest:
```

```
01_MetroverseWireframeQuad] Started
[...]
```

2. Running at a specific resolution:

> `.\SPECviewperf-CLI.exe -w maya-07` // Not adding `--resolution` param will default to 4K resolution, if supported by the system.

> `.\SPECviewperf-CLI.exe -w maya-07 --resolution FullHD` // Will run at FullHD resolution, if supported by the system.

3. Debug Logs:

> `.\SPECviewperf-CLI.exe -w maya-07 --resolution FullHD --debug`

Output:

```
{
  iterations: 1,
  workloadGap: 0,
  resolution: 'FullHD',
  path: 'viewsets/',
  workload: [ 'maya-07' ],
  debug: true
}
[2025-01-08|09:35:51] Gathering System Details: Started
[2025-01-08|09:35:56] Gathering System Details: Finished
[...]
```

Results and Reporting

Scoring

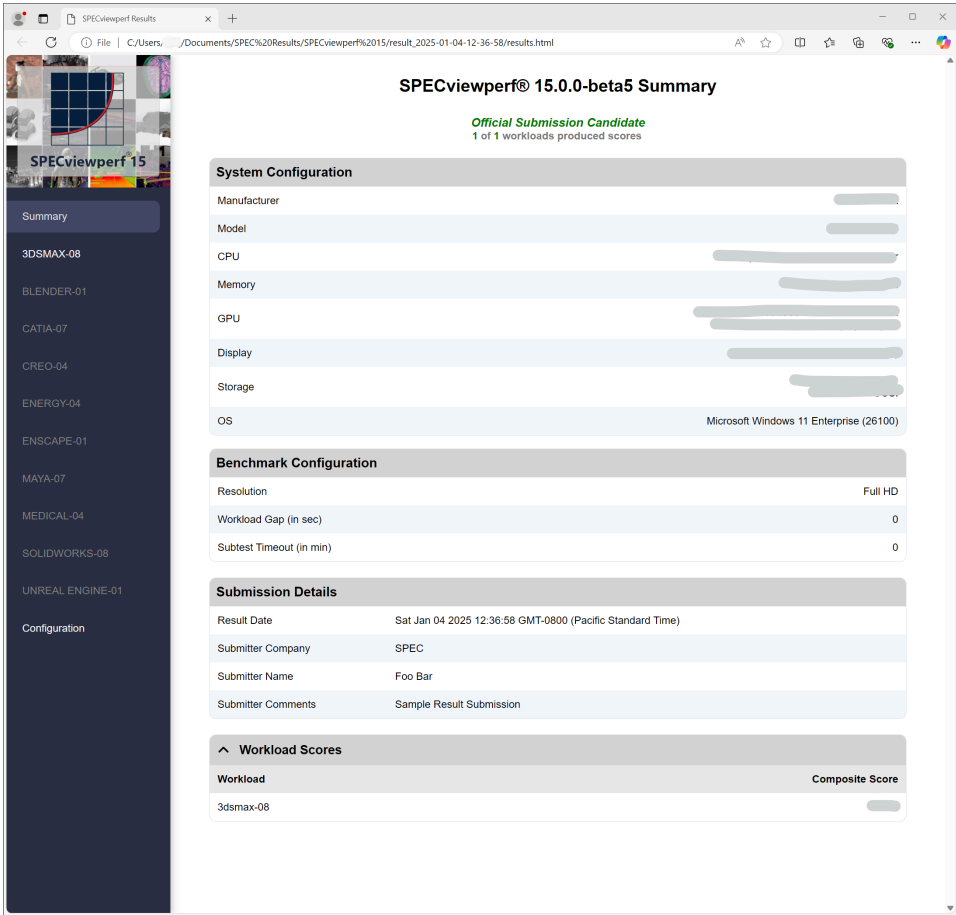
The benchmark reports two types of scores:

- *Subtest scores* are in "Frames per Second".
- *Workload composite scores* are calculated by taking a weighted geometric mean of the constituent subtest scores (individual subtest weights are available in the CSV result output.)

Result Output

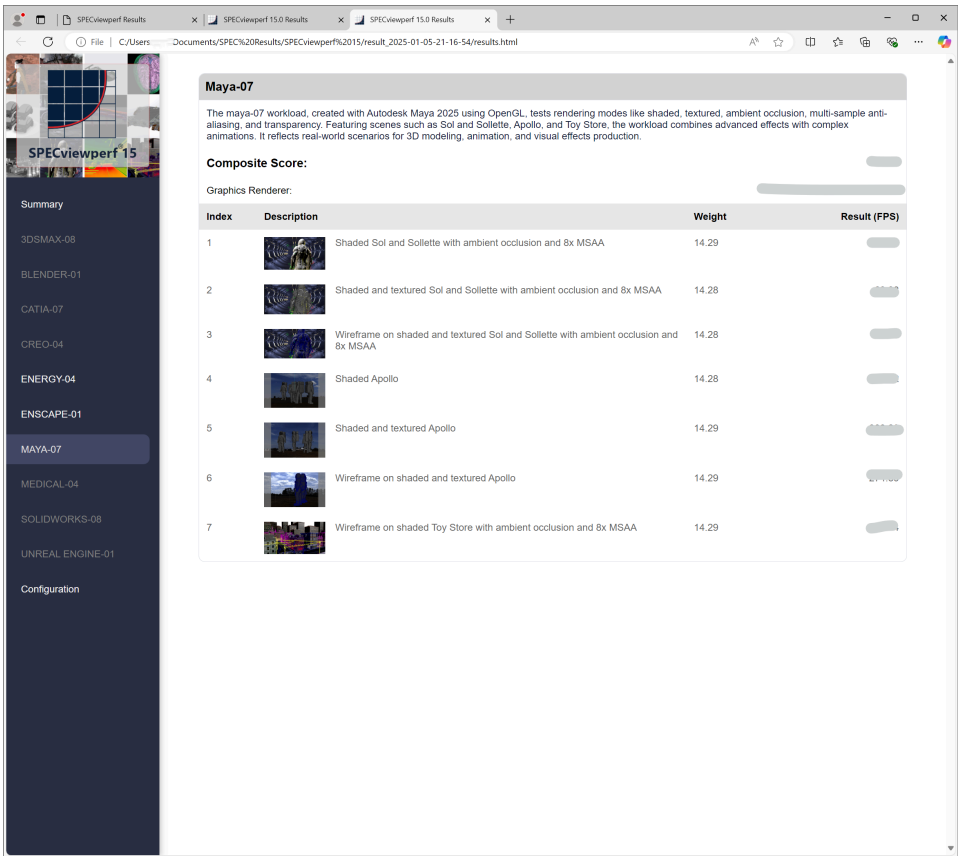
Each benchmark run and iteration produces a result folder with scores reported in HTML, CSV, and JSON formats.

The **HTML result summary page** shows select [system configuration](#) information, [benchmark configuration](#) information, [submission details](#) (if entered by the user), and all calculated composite category and workload SPEC ratios. If any subtest of a category or workload did not produce a score, either because it was unselected or encountered errors, the composite score will be omitted from the results.



HTML results summary page

Users may view more details for a workload by selecting its page in the left navigation bar. Individual workload pages list all contributing subtests. The **Workload Scores** page shows the full data for every workload and subtest that completed successfully.



HTML results workload page

Comparing Results Across Different Systems

Users may compare their system performance to results posted by SPEC on the official [SPECviewperf Results](#) page.

Note: 4K and Full HD runs of any workload cannot be compared directly. Official results are presented in separate tables for each resolution.

Software Updates

- An update check is performed at benchmark startup; if an update is available, a tooltip will be shown at the top of the [main window](#).
- The [benchmark configuration](#) window will facilitate upgrading a workload if a new version is available.

An internet connection is required for any updates.

Version History

This section documents the release history and changes across different versions of SPECviewperf 15.

Version	Release Date	Updates	Notes
15.0.1	Dec 11, 2025	• Adds new workload: snx-05 • Minor bug fixes	• Results from version 15.0.1 are still comparable to version 15.0.0. Previous workloads have not been modified.

Version	Release Date	Updates	Notes
			• Users running the full benchmark suite should update to this version to include the new workload in their results.
15.0.0	May 1, 2025	• Official release of SPECviewperf 15.0.0 with 10 workloads: 3dsmax-08, blender-01, catia-07, creo-04, energy-04, enscape-01, maya-07, medical-04, solidworks-08, unreal_engine-01 • Enhanced usability with a redesigned graphical user interface (GUI) and streamlined installation and configuration processes	Initial release

Uninstalling SPECviewperf

- Individual workloads may be uninstalled from the [benchmark configuration](#) window.
- The benchmark and all workloads may be uninstalled from the Windows "Add or remove programs" control panel interface.
- Result folders will not be erased when uninstalling, but can be safely deleted manually (default location for results is %USERPROFILE%\Documents\SPEC Results\SPECviewperf 15).

Benchmark Compliance

Run Rules

1. The system under test must meet all [minimum requirements](#) as listed in the SPECviewperf user guide
2. The system under test must perform all the respective graphics API's functionality and other work requested by the benchmark.
3. The system must be conformant for the pixel format or visual used by each API (OpenGL, Vulkan, DirectX).
4. Settings for environment variables, registry variables and hints must not disable compliant behavior.
5. No interaction is allowed with the system under test during the benchmark, unless required by the benchmark.
6. The system under test cannot skip frames during the benchmark run.
7. It is not permissible to change the system configuration during the running of a given benchmark. For example, one cannot power off the system, make some changes, then power back on and run the rest of the benchmark.
8. The system should have at least 16 GB of physical memory.
9. Screen grabs for SPECviewperf will be full window size.
10. The color depth used must be at least 24 bits (true color), with at least 8 bits of red, 8 bits of green and 8 bits of blue.
11. If a depth buffer is requested, it must have at least 24 bits of resolution.
12. The display resolution of the main display must match or exceed the selected resolution in the SPECviewperf UI (i.e. 1920x1080 for Full HD or 3840x2160 for 4K.)

13. The display resolution of the main display must be large enough to run the individual tests at their requested window size, with no reduction or clipping of test window contents.
14. Screen DPI scaling must be set to 100% and the entire viewport of a subtest must remain fully visible on-screen for each workload. Results which show viewport warnings in workload log files are unacceptable for publication.
15. No UI or window elements, including but not limited to taskbars, toolbars, menubars, and titlebars, may obstruct the viewport rendering context, except for the SPECviewperf progress GUI. These elements must be configured to auto-hide or be resized so that all test window contents are fully visible on-screen without being clipped or obscured.
16. Tests may be run with or without a desktop/window manager but must be run on some native windowing system.
17. Results to be made public must be generated by the official benchmark which may not be changed, except to correct errors in reporting by the benchmark, and these changes shall be disclosed voluntarily upon submission.
18. Virtualized configurations, defined as any operating system configuration running on a hypervisor or virtualization layer of any kind, must include the word "virtualized" in the comment field of the config.txt file. This information must be populated before the execution of the benchmark to ensure the results reflect this attribute.
19. Virtualized configurations as defined above must also include a declaration of the transport layer name and version used by the virtualization software in parenthesis after the graphics accelerator listed in the Graphics Accelerator field of the Graphics Hardware Configuration section of the result file.
20. Virtualized configurations as defined above must also include a declaration of the hypervisor name and version used by the virtualization software in parenthesis after the model listed in the Model field of the System Hardware Configuration section of the result file.
21. All configurations shall include a link (URL) to a publicly accessible device driver package used to generate the submission.

Submission Candidacy

A submission candidate is a result that may be submitted to SPEC for publication on the official [SPECviewperf results](#) page. The following are the criteria for a submission candidate to be published by SPEC:

1. The user follows the [run rules](#) when collecting results.
2. The machine model and configuration is available for purchase by the general public at the time of submission.
3. *At least one* official workload completes without errors or warnings.
4. In a multi-GPU configuration, all enabled workloads must utilize the same graphics renderer.
5. The result is [packaged with complete submission details](#).

The results manager of the benchmark will automatically mark a result as a submission candidate if all workloads completed successfully, but it cannot validate all the above criteria. If submitted to SPEC, a result will be reviewed for conformance to the submission criteria.

Any unmet criteria will disqualify the result from being a submission candidate.

Information on where to submit results and the associated costs are detailed on the [SPECgwpwg membership page](#).

Technical Support

Known Issues

Review the known issues and workarounds if you are encountering problems in running the benchmark or are generating unexpected results.

General Issues

1. If any crashes or incorrect rendering are encountered, it is recommended to update to the latest GPU-vendor-provided graphics driver.
2. Using the [Command-Line Interface](#) in Windows 11 24H2 or newer may result in reduced performance or timeouts if the operating system relegates `SPECviewperf-CLI.exe` to a background task. Running via the [GUI](#) and keeping the benchmark in focus is a workaround.
3. Systems may experience thermal saturation during a benchmark run, increasing variability and reducing performance. Improve system cooling or thermals, or workaround thermal limits by increasing the workload gap in [Advanced Settings](#).
4. Results folder may contain `queryLog` folders that were not deleted automatically. These may be deleted safely if not being used for debugging a benchmark initialization failure.
5. Calculated [Composite Score](#) values may differ slightly between the JSON and the CSV/HTML result files due to differences in precision and rounding. Official scores should be reported from the CSV or HTML result files.
6. Some machine configurations (typically machines with specifications below the [system requirements](#)) may timeout while running workloads. Timeouts are disabled by default and can be set in the [Advanced Settings](#) menu or via a [CLI option](#). Results may still be considered [submission candidates](#) if timeouts are disabled.

Windows 10 Support

Windows 10 may run but is not officially supported. The following issues are known:

- High core-count (i.e. >32 cores) or heterogenous core configuration (e.g. performance/efficiency or classic/dense) systems may encounter low performance. No known workaround, caused by limited support for high core count or heterogeneous core configurations on Windows 10.

Common Installation and Configuration Issues

- File corruption can result in verification failures which can fail installation of a workload.
- Modification of workloads after installation can result in workload being disabled from benchmark run.
- After installation, you may encounter dialog popups requesting permission for Windows and apps to access your location. It is acceptable to decline these requests by selecting `No`.

Workload Issues

- *3dsmax-08*
 - The Minerva subtests (06_MinervaShadedEdges, 09_MinervaShaded & 10_MinervaShadedHQ) may show an incorrect right edge in FullHD. Users running on Intel or NVIDIA hardware may see slight differences in the right edge of the rendered frames.
- *catia-07*

- The wings in Subtest 04_ThreeLoftJetsWhite look corrupted. This is not a bug in the test, but an artifact in the model source.
- In 3DEXPERIENCE, the SSAO algorithm does not handle buffer rescaling, which has led to some issues with offscreen 4K rendering appearing in 2K traces. In those tests, the frames were post-processed to correct this.
- In 3DEXPERIENCE tests, some minor run-to-run variance may occur resulting in non-deterministic screen captures.
- *creo-04*
 - Subtest 07_WorldCarShadedEdges may show flickering effect coming from z-fighting at the bottom of the car.
- *maya-07*
 - Subtests 04_ApolloShaded, 05_ApolloShadedTextured, and 06_ApolloWireframeShadedTextured utilize 8x MSAA, although this antialiasing setting is not documented in the workload description.
 - Subtest 07_ToyStoreWireframeShaded has a z-bleeding visual artifact (seen in textures of the building) that causes a flickering effect. The source of the issue is in the model being used and this is how Maya application renders it.
- *medical-04*
 - The 'BeatingHeart' tests may produce non-deterministic screen grabs.
- *unreal-engine-01*
 - The City Sample project may experience extended loading times when first opened, introducing risk of the benchmark starting before all resources are fully loaded. To mitigate this issue, the workload waits for 60 seconds after opening the project before starting measurements. It is also recommended to review the generated screenshots ([Screenshots for reference](#)), to ensure all resources were loaded.
 - [Distance Fields](#) are disabled by default for integrated graphics on UE 5.4. We have applied a patch from UE 5.5 that enables them for newer iGPUs.

Contacting Support

Reach out to us via the [SPECgwpwg support page](#).

FAQs

What is SPECviewperf?

The SPECviewperf benchmark is a performance evaluation tool created to test professional workstation use cases. Refer to the [Benchmark Introduction](#) section for more information.

Who develops SPECviewperf?

The SPECviewperf benchmark is developed by the [SPEC Graphics & Workstation Performance Group \(SPECgwpwg\)](#), specifically its SPEC Graphics Performance Characterization (SPECgpc) subcommittee.

How many workloads and tests are included?

SPECviewperf 15.0.1 is released with 11 official workloads with a total of 93 subtests. Inspect an [official result](#) for a full list of tests in the benchmark.

How do I download the benchmark?

Download it from the [SPECgwpwg website \[Direct Link\]](#). It is provided free for everyone except sellers of computers and related products.

How do I create an offline installation package?

Refer to [Offline Install](#) for instructions to create an offline installer.

Where is the benchmark installed?

It is installed in `Program Files\SPECviewperf 15` by default. Users may change the default install folder during setup.

What are the minimum system requirements to install and run the benchmark?

Refer to [System Requirements](#) for the minimum supported system configuration.

Can I install or run with only a specific set of workloads?

Yes, you can install a subset of workloads during the second step of [installation](#). Each benchmark run can be [configured](#) to run a subset of installed workloads.

Can I configure the number of benchmark iterations?

Yes, the number of iterations can be configured in the [main window](#). Note that each iteration will generate a separate result folder. Results from multiple iterations are not automatically combined or averaged.

My display is set to 4K, but the 4K option is still greyed out in the benchmark window. Why?

Verify if the display set to 4K is your [main display](#). The benchmark checks the resolution of your primary display. You can set your main display in Settings > System > Display under [Multiple displays](#).

I've changed my display DPI settings. But the benchmark GUI still shows the old value. Why?

For some internal Windows settings, including changes to the DPI, to be fully applied, the system needs to be signed out and then signed back in. After this, the benchmark will accurately read and reflect the updated DPI value.

How can I automate or script benchmark runs?

Refer to the [Command-Line Interface](#) section for details on how to access and run from a console or scripts. The CSV output may also be used for automated result parsing.

Note We do not recommend using SPECviewperf 15 as a stress test, and any issues that arise from such usage will not be addressed by the SPEC Graphics and Workstation Performance Group (GWPG).

Can I disable certain subtests of a workload?

No, All the subtests must be run to complete any workload execution.

How are the benchmark results scored?

Refer to the [Scoring](#) sections for details on results and scores.

Where are the benchmark results stored?

The default result folder is %USERPROFILE%\Documents\SPEC Results\SPECviewperf 15; however the user may change the folder in the [advanced settings](#) menu.

How are results output and formatted?

Results are output in JSON, CSV, and HTML formats with snapshots of each selected workload for every run.

How can I view the benchmark results?

Results may be viewed from the [Manage Results](#) window in the GUI or by navigating directly to the default result folder: %USERPROFILE%\Documents\SPEC Results\SPECviewperf 15.

How do I ensure my benchmark run qualifies for submission and publication by SPEC?

Refer to the [Submission Candidate](#) section for submission criteria.

What happens if my submission is disqualified?

Submissions not meeting the [submission criteria](#) may be disqualified and returned to the submitter with comments or suggestions. Note that submission fees are only collected for published results, so a disqualified submission will not incur any costs.

How do I package my results for submission?

Refer to the [Result Packaging and Submission](#) section for details on how to prepare a package for submission.

How do I use the Command-Line interface (CLI)?

Refer to the [Command-line interface](#) section for how to invoke the command-line interface and display useful parameters and options.

Can I run individual workloads using CLI?

Yes, the CLI can launch individual workloads. Refer to the [Command-line interface](#) section for more details.

What is the workload gap setting?

A workload gap sets a delay between workload executions. Refer to the [Advanced Settings](#) section for more details about this option.

What are the known issues or bugs?

Refer to the [Known Issues](#) section to find a list of known problems and some workarounds.

What should I do if the benchmark triggers a virus scan?

Windows Real-time protection or other virus scans may impact performance if they trigger during a benchmark run. Users may temporarily disable virus scanners or setup virus scanner exclusions at their own risk.

What is a "Candidate" result?

Refer to the [Submission Candidate](#) section for more information.

Is it possible to uninstall workloads but keep the results?

Yes, uninstalling individual workloads or the entire benchmark does not delete any results.

Unreal Engine does not satisfy SM6_Atomics requirement. Why?

You can check the DirectX capabilities supported by your hardware and driver using [DirectX Capabilities Viewer](#) to learn more.

What should I do if some workloads fail during the benchmark run?

Ensure your machine meets the [system requirements](#). Refer to the [Known Issues](#) section to see if your problem is documented and has a workaround. Check the HTML results for information on the errors. Check the output and error logs inside the results directory. If the error persists, consider submitting a bug report with your result folder and any relevant logs/details on the [SPECgwpkg support page](#).

How do I report issues or get help?

See if your question or issue is answered by this user guide or documented in the [Known Issues](#) section. If you need further assistance reach out to us via the [SPECgwpkg support page](#).

Appendix

Glossary of Terms

Term	Definition
Workload	A set of graphics subtests representing actual graphics API use by the given application
Subtest	A subset of the workload testing a specific model and rendering mode for the given application

Reference Documents and Links

1. [Standard Performance Evaluation Corporation \(SPEC\)](#)
2. [SPEC Graphics & Workstation Performance Group \(SPECgwpwg\)](#)
3. [Official SPECviewperf Results](#)
4. [Reference Images](#)

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3dsmax

Various 3ds Max models were contributed by:

- *Autodesk*
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Blender

- Classroom Scene's Original Author: *Christophe Seux*
- The Metroverse scene was contributed by *Intel*

Catia

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Creo

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- The Scorpion 1 vehicle is provided courtesy of *Charlie Borg via GrabCad*

Energy

The seismic volumes used by the energy viewset were derived from various files found at https://wiki.seg.org/wiki/Open_data.

- Blake Ridge
 - URL: <https://terranubis.com/datainfo/Blake-Ridge-Hydrates-3D>

- F3 Netherlands
 - URL: <https://terranubis.com/datainfo/Netherlands-Offshore-F3-Block-Complete>
- Opunake
 - URL: <https://wiki.seg.org/wiki/Opunake-3D>
 - *New Zealand Petroleum and Minerals (NZPM)*

Enscape

- The Villa ("Revit Villa") and Courtyard ("Rhino Interior") scenes are provided courtesy of *Chaos*
- Amazon Lumberyard Bistro
 - Author: *Amazon Lumberyard*
 - Title: *Amazon Lumberyard Bistro, Open Research Content Archive (ORCA)*
 - URL: <http://developer.nvidia.com/orca/amazon-lumberyard-bistro>
- San Miguel 2.0
 - Author: © *Guillermo M. Leal Llaguno*
 - URL: <http://www.evvisual.com/>

Maya

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Medical

- Human chest animation: *Lucile Packard Children's Hospital Lucas Center for Imaging, Stanford University School of Medicine*
- Stag beetle: *Georg Glaeser, Vienna University of Applied Arts, Austria; scanned by Johannes Kastner, Wels College of Engineering, Austria, and Eduard Gröller, Vienna University of Technology, Austria*
- Human head and torso: *Provided by a member of the SPECgpc subcommittee*
- Alligator head: *William Ruger Porter of Ohio University, Jay C. Sedlmayr of Louisiana State University, and Lawrence M. Witmer of Ohio University via Dryad Dataset (licensed under CC0 1.0 Universal Public Domain Dedication)*

NX

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Solidworks

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- Other models are provided courtesy of *Dassault Systemes*

Unreal Engine

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